Lecture 16: Stability of LU with Pivoting; Interpolation

Math 6610

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Today's Goals

- Finish Lecture 15: $\kappa(A)$ as stretching, distance to singular, changes in b
- See why LU without pivoting is not backward stable
- Pose the interpolation problem; prove existence and uniqueness (Lagrange)

Review of Lecture 15

Remark (*Setting*). $A \in \mathbb{C}^{n\times n}$ nonsingular (invertible), $b \neq 0$, $x = A^{-1}b$. $\|\cdot\|$: a vector norm on $\mathbb{C}^n$ and its lub norm on matrices.

Remark (*Condition number*). $\kappa(A) := \|A\| \left\|A^{-1}\right\| \geqslant 1$, and for $(A + \delta A)(x + \delta x) = b$,

$$\frac{\|\delta x\|}{\|x\|} \leqslant \frac{\kappa(A)}{1 - \|A^{-1}\| \, \|\delta A\|} \frac{\|\delta A\|}{\|A\|} \qquad \text{whenever } \left\|A^{-1}\right\| \|\delta A\| < 1,$$

and no constant smaller than $\kappa(A)$ works as $\delta A \to 0$.

Remark (*Accuracy theorem, Lecture 14*). Backward stable algorithm $\Rightarrow$ relative output error $\leqslant 2\kappa \cdot C\,\epsilon_{\text{mach}}$ for small ϵ_{mach}.

Remark (*Today*). Is LU (Gaussian elimination) backward stable? Then: interpolation.

$\kappa(A)$ as Stretching

Proposition. For nonsingular A and any lub norm,

$$\|A^{-1}\| = \frac{1}{\min_{\|z\|=1} \|Az\|}, \qquad \kappa(A) = \frac{\max_{\|z\|=1} \|Az\|}{\min_{\|z\|=1} \|Az\|} :$$

largest stretch of A divided by its smallest stretch.

Proof. Idea: A^{-1} undoes A; substitute $y = Az$.
1. The minimum exists and is > 0. $\|Az\|$ is continuous on the closed, bounded set $\{\|z\| = 1\}$, and $Az \neq 0$ there.
2. Substitute $y = Az$ ($y \neq 0 \Leftrightarrow z \neq 0$, as A is invertible):

$$\|A^{-1}\| = \max_{y \neq 0} \frac{\|A^{-1}y\|}{\|y\|} = \max_{z \neq 0} \frac{\|z\|}{\|Az\|} = \frac{1}{\min_{\|z\|=1} \|Az\|}.$$

3. Multiply by $\|A\| = \max_{\|z\|=1} \|Az\|$. ■

$\kappa(A)$ in the 2-Norm; Distance to Singular

Remark (*2-norm*). $\sigma_{\max} := \max_{\|z\|_2=1} \|Az\|_2$ and $\sigma_{\min} := \min_{\|z\|_2=1} \|Az\|_2$ are the largest and smallest singular values of A:

$$\kappa_2(A) = \sigma_{\max}/\sigma_{\min}.$$

$\kappa_2(A) = 1$ exactly when $A = cQ$, $c \neq 0$, Q unitary: all stretches equal, the unit sphere goes to a sphere. ("If": $\|cQz\|_2 = |c| \, \|z\|_2$. "Only if": not shown.)

Remark (*Distance to the nearest singular matrix*). For any lub norm,

$$\min\left\{ \frac{\|\delta A\|}{\|A\|} : A + \delta A \text{ singular} \right\} = \frac{1}{\kappa(A)}.$$

"$\geqslant$" is Step 1 of the proof in Lecture 15; "$=$" uses a δA as in Step 4 there (not shown). Large $\kappa(A) \Leftrightarrow A$ is relatively close to a singular matrix.

Changing b Instead of A

Proposition. Let A be nonsingular, $b \neq 0$, $Ax = b$ and $A(x + \delta x) = b + \delta b$. Then (exactly, no approximation)

$$\frac{\|\delta x\|}{\|x\|} \leqslant \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$

Proof. Idea: bound the numerator and the denominator separately.

1. *Numerator.* Subtracting gives $A\,\delta x = \delta b$, so $\|\delta x\| \leqslant \left\|A^{-1}\right\| \|\delta b\|$.
2. *Denominator.* $\|b\| = \|Ax\| \leqslant \|A\| \|x\|$, so $\dfrac{1}{\|x\|} \leqslant \dfrac{\|A\|}{\|b\|}$.
3. *Multiply* the two bounds:

$$\frac{\|\delta x\|}{\|x\|} \leqslant \left\|A^{-1}\right\| \|A\| \frac{\|\delta b\|}{\|b\|} = \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$

■

Remark. Changes in A and in b are both magnified by at most about $\kappa(A)$.

LU Without Pivoting: Backward Error

Theorem (*Proof omitted*). Let $A \in \mathbb{C}^{n\times n}$ be nonsingular with $A = LU$ (L lower triangular with ones on the diagonal, U upper triangular; i.e. elimination without row swaps meets no zero pivot). For all small enough ϵ_{mach}, LU without pivoting in floating point runs to the end and returns $\tilde{L}$, $\tilde{U}$ with

$$\tilde{L}\tilde{U} = A + \delta A, \qquad \frac{\|\delta A\|}{\|L\|\ \|U\|} = O(\epsilon_{\text{mach}}).$$

Remark (*Meaning of $O(\epsilon_{\text{mach}})$*). $\|\delta A\| \leqslant C\,\epsilon_{\text{mach}}\,\|L\|\,\|U\|$ for small ϵ_{mach}, with C depending on n but not on A.

Remark (*When is this good?*). If $\|L\|\,\|U\| \leqslant c\ \|A\|$ with c independent of A, then $\|\delta A\| \,/\, \|A\| \leqslant Cc\,\epsilon_{\text{mach}}$: backward stable. Without pivoting no such c exists, and the example shows δA really can be large.

LU Without Pivoting Fails: Example (1/2)

Example (*Double precision*, $\eta := 2^{-66} \approx 1.4 \cdot 10^{-20}$).

$$A = \begin{bmatrix} \eta & 1 \\ 1 & 1 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 \\ \eta^{-1} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} \eta & 1 \\ 0 & 1 - \eta^{-1} \end{bmatrix}.$$

The multiplier $\eta^{-1} = 2^{66}$ is exact. Near 2^{66} doubles are 2^{13} apart, so $1 - 2^{66}$ rounds to -2^{66}:

$$\tilde{U} = \begin{bmatrix} \eta & 1 \\ 0 & -2^{66} \end{bmatrix}, \qquad \tilde{L}\tilde{U} = \begin{bmatrix} \eta & 1 \\ 1 & 0 \end{bmatrix}, \qquad \delta A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}.$$

(Entry $(2,2)$: $2^{66} \cdot 1 + (-2^{66}) = 0$.) In the ∞-norm (max row sum),

$$\frac{\|\delta A\|_\infty}{\|A\|_\infty} = \frac{1}{2}, \qquad \|L\|_\infty = 2^{66} + 1, \qquad \|U\|_\infty = 2^{66} - 1.$$

LU Without Pivoting Fails: Example (2/2)

Example (*Solving with $b = (1,2)^T$*). Exact: $x = \left(\frac{1}{1-\eta}, \frac{1-2\eta}{1-\eta}\right)^T \approx (1,1)^T$. Computed with $\tilde{L}$, $\tilde{U}$:

$$\tilde{y} = \begin{bmatrix} 1 \\ \mathrm{fl}(2 - 2^{66}) \end{bmatrix} = \begin{bmatrix} 1 \\ -2^{66} \end{bmatrix}, \qquad \tilde{x}_2 = 1, \quad \tilde{x}_1 = \frac{1-1}{\eta} = 0.$$

So $\tilde{x} = (0,1)^T$: no correct digit in $\tilde{x}_1$. Yet $\kappa_\infty(A) = 2 \cdot \frac{2}{1-\eta} \approx 4$ (check: $A^{-1} = \frac{1}{\eta - 1}\begin{bmatrix} 1 & -1 \\ -1 & \eta \end{bmatrix}$).

Remark (*Not backward stable, not even stable*). A backward stable method would give error $\leqslant 2 \cdot 4 \cdot C\,\epsilon_{\text{mach}}$ (accuracy theorem). And every matrix within relative distance $O(\epsilon_{\text{mach}})$ of A has its solution within $O(\epsilon_{\text{mach}})$ of $(1,1)^T$ (bound in the review), far from $\tilde{x}$: not stable either.

Partial Pivoting and the Growth Factor

Definition (*Partial pivoting*). At elimination step k, swap row k with a row $i \geqslant k$ whose entry in column k is largest in absolute value (ties: the smallest such i, as in MATLAB/LAPACK). Result: $PA = LU$, with P a permutation matrix (the identity with its rows reordered).

Remark (*Effect on L*). Each multiplier is (entry)/(pivot), and the pivot is the largest entry:

$$\left|l_{jk}\right| \leqslant 1 \quad \text{for all } j,k \qquad \Longrightarrow \qquad \|L\|_\infty \leqslant n.$$

What about U?

Definition (*Growth factor*). With $PA = LU$ from partial pivoting in exact arithmetic,

$$\rho = \rho(A) := \frac{\max_{j,k} \left|u_{jk}\right|}{\max_{j,k} \left|a_{jk}\right|} :$$

how much the entries grow during elimination.

LU With Partial Pivoting: Backward Error

Theorem (*Proof omitted*). Let $A \in \mathbb{C}^{n\times n}$ be nonsingular. LU with partial pivoting in floating point returns $\tilde{L}$, $\tilde{U}$ and a permutation $\tilde{P}$ with

$$\tilde{L}\tilde{U} = \tilde{P}A + \delta A, \qquad \frac{\|\delta A\|}{\|A\|} = O(\rho\, \epsilon_{\text{mach}}),$$

i.e. $\|\delta A\| \leqslant C\rho\, \epsilon_{\text{mach}}\, \|A\|$ for small ϵ_{mach}, C depending on n only.

Remark (*The example again*). Swap rows: $\tilde{P}A = \left[\begin{smallmatrix} 1 & 1 \\ \eta & 1 \end{smallmatrix}\right]$, multiplier η, and $\text{fl}(1-\eta) = 1$ ($\eta < \frac{1}{2}\epsilon_{\text{mach}}$):

$$\tilde{L}\tilde{U} = \begin{bmatrix} 1 & 0 \\ \eta & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \eta & 1+\eta \end{bmatrix}, \qquad \frac{\|\delta A\|_\infty}{\|A\|_\infty} = \frac{\eta}{2}.$$

How Large Can $\rho(A)$ Get?

Proposition. $\rho(A) \leqslant 2^{n-1}$ for every nonsingular $A \in \mathbb{C}^{n\times n}$.

Proof. Idea: with $|l_{ik}| \leqslant 1$, one step at most doubles the largest entry. Let μ_k := largest |entry| before step k ($\mu_1 = \max |a_{jk}|$: swaps only reorder entries).

1. One step. Rows $i > k$ change to

$$\left|a_{ij}^{(k)} - l_{ik}\, a_{kj}^{(k)}\right| \leqslant \mu_k + 1 \cdot \mu_k \quad \Longrightarrow \quad \mu_{k+1} \leqslant 2\mu_k.$$

2. Induction on k. $\mu_k \leqslant 2^{k-1}\mu_1$; after the last step ($k = n$) the matrix is U: $\max |u_{jk}| \leqslant 2^{n-1} \max |a_{jk}|$. ■

The Worst Case: $\rho = 2^{n-1}$ Is Attained

Proposition. Let $A \in \mathbb{R}^{n\times n}$ have 1 on the diagonal, -1 below it, 1 in the last column, 0 elsewhere. Partial pivoting makes no row swaps and gives

$$A = \begin{bmatrix} 1 & 0 & \cdots & 0 & 1 \\ -1 & 1 & \ddots & \vdots & 1 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ -1 & \cdots & -1 & 1 & 1 \\ -1 & \cdots & -1 & -1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \ddots & \vdots & 2 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & 2^{n-2} \\ 0 & \cdots & 0 & 0 & 2^{n-1} \end{bmatrix},$$

$L =$ (1 on the diagonal, -1 below it). Hence $\rho(A) = 2^{n-1}/1 = 2^{n-1}$.

Remark (*Picture*). Only the last column changes; each step doubles it below the pivot row: $(1, 1, \ldots, 1)^T \to (1, 2, 2, \ldots, 2)^T \to (1, 2, 4, \ldots, 4)^T \to \cdots$

The Worst Case: Proof

Proof. Idea: each step adds the pivot row to every row below. The pivot row has only two nonzeros (1 on the diagonal, a power of 2 in column n), so only the last column changes, and it doubles.
Claim (induction on k): after k steps, without swaps, each row below row k still has its original entries in columns $k+1, \ldots, n-1$ and has 2^k in column n.
Base $k = 0$: this is A.
1. No swap. Column $k+1$ below the diagonal: 1, then all -1. Tie $\Rightarrow$ pivot row $k+1$; multipliers $l_{i,k+1} = -1$.
2. Add row $k+1$ (entries 1 and 2^k) to each row $i > k+1$:

$$(i, k+1): \ -1+1=0, \qquad (i, n): \ 2^k + 2^k = 2^{k+1}.$$

3. End ($k = n-1$). $u_{nn} = 2^{n-1}$, all $\left|u_{jk}\right| \leqslant 2^{n-1}$, and max $\left|a_{jk}\right| = 1$. ■

The Worst Case in Floating Point

Remark (*The problem is well conditioned*). For this A: $\kappa_\infty(A) = n$ and $\kappa_2(A) \approx 28.6$ for $n = 64$ (computed numerically). So any error comes from the algorithm.

Remark (*Experiment: double precision, MATLAB `A\b` / LAPACK*). Take x with random normal entries, $b := Ax$, solve with LU + partial pivoting:

n	8	32	64
$\rho(A) = 2^{n-1}$	128	$2.1 \cdot 10^9$	$9.2 \cdot 10^{18}$
$\|\tilde{x} - x\|_\infty / \|x\|_\infty$	$\approx 10^{-15}$	$\approx 10^{-8}$	≈ 0.7

At $n = 64$: no correct digits, although $\kappa_\infty(A) = 64$.

Remark (*Typical matrices*). Same experiment with random normal A: we observed $\rho \approx 3$–7 ($n = 64$), 8–11 ($n = 256$), 15–19 ($n = 1024$).

What the Growth Bound Means

Remark (*Formally backward stable*). 2^{n-1} depends on n only. So, by the theorem, for each fixed n,

$$\frac{\|\delta A\|}{\|A\|} \leqslant C\, 2^{n-1} \epsilon_{\text{mach}} \quad \text{for all nonsingular } A \in \mathbb{C}^{n\times n}:$$

LU with partial pivoting is backward stable in the sense of Lecture 14 (the constant may depend on n).

Remark (*But the constant is useless*). By the worst-case matrix, no bound $C\, p(n)\, \epsilon_{\text{mach}}$ with a polynomial $p(n)$ holds for all A. For $n = 64$ the guaranteed bound is at least $2^{63}\epsilon_{\text{mach}} \geqslant 10^3$: it says nothing.

Remark (*In practice*). For matrices met in applications $\rho(A)$ is small (a modest function of n), and $A\backslash b$ is accurate whenever $\kappa(A)\epsilon_{\text{mach}} \ll 1$. Large growth from real applications is rare but known (Wright 1993: boundary value problems; Foster 1994: integral equations).

Putting the LU Story Together

Everything depends on how $\|L\|\,\|U\|$ compares with $\|A\|$.

- **No pivoting:** $\|L\|$ can be arbitrarily large (2^{66} in the example) and δA as large as A itself: not stable.
- **Partial pivoting:** in the ∞-norm, using $\max \left|a_{jk}\right| \leqslant \|A\|_\infty$,

$$\|L\|_\infty \leqslant n, \qquad \|U\|_\infty \leqslant n \max \left|u_{jk}\right| = n\rho \max \left|a_{jk}\right| \leqslant n\rho \ \|A\|_\infty ,$$

so $\|L\|_\infty \|U\|_\infty \leqslant n^2 \rho \ \|A\|_\infty$: this is where $O(\rho\, \epsilon_{\text{mach}})$ comes from.

Remark (*Accuracy, preview*). The two triangular solves add backward errors of the same size (proof omitted). The accuracy theorem (Lecture 14) then gives

$$\frac{\|\tilde{x} - x\|}{\|x\|} = O\big(\kappa(A)\, \rho(A)\, \epsilon_{\text{mach}}\big) :$$

problem (κ) $\times$ growth (ρ) $\times$ rounding (ϵ_{mach}).

Interpolation: The Problem

Definition (*Interpolation*). Given $n+1$ data points

$$(x_j, y_j), \qquad j = 0, 1, \ldots, n, \qquad x_0 < x_1 < \cdots < x_n, \tag{1}$$

with $x_j, y_j \in \mathbb{R}$, a function p *interpolates* the data if $p(x_j) = y_j$ for $j = 0, 1, \ldots, n$. The x_j are the *nodes*.

Remark (*Where it is used*).

- Values between measurements: readings every hour, needed at 2:15.
- Resampling: resizing images, changing audio sample rates (MATLAB `interp1`, `interp2`).
- Inside other methods: integrate or differentiate the interpolant (quadrature, Lecture 23; ODE solvers).

Remark (*Wanted*). A "simple" p: cheap to evaluate, differentiate and integrate. First choice: polynomials. Later: piecewise polynomials (splines).

Polynomial Interpolation: A Linear System

Definition (*Polynomials of degree $\leqslant n$*).

$$\prod_n := \left\{ P(x) = c_0 + c_1 x + \cdots + c_n x^n \;\middle|\; c_0, c_1, \ldots, c_n \in \mathbb{R} \right\}$$

= set of all (real) polynomials of degree $\leqslant n$: $n+1$ coefficients.

Remark (*Monomial basis*). $P(x_j) = y_j$, $j = 0, \ldots, n$, are $n+1$ linear equations for the $n+1$ coefficients:

$$\underbrace{\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix}}_{=: V \text{ (Vandermonde matrix)}} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

Square system: solvable by LU with partial pivoting. Is V nonsingular? How well conditioned?

Lagrange Basis and Existence/Uniqueness

Definition (*Lagrange basis polynomials*). For distinct nodes $x_0, \ldots, x_n$ and $j = 0, \ldots, n$,

$$\ell_j(x) := \prod_{\substack{k=0 \\ k \neq j}}^{n} \frac{x - x_k}{x_j - x_k}.$$

Theorem (*Polynomial interpolation*). Let $x_0 < x_1 < \cdots < x_n$ and $y_0, \ldots, y_n \in \mathbb{R}$. There is exactly one $P_n \in \prod_n$ that interpolates the data (1):

$$P_n(x) = \sum_{j=0}^{n} y_j \, \ell_j(x) \qquad \text{(Lagrange interpolation formula)}.$$

Equivalently, the Vandermonde matrix V is nonsingular.

Polynomial Interpolation: Proof

Proof. Idea: ℓ_j is 1 at x_j and 0 at the other nodes; uniqueness by counting zeros.
1. $\ell_j \in \prod_n$ *and* $\ell_j(x_l) = \delta_{jl}$ (1 if $l = j$, else 0). n linear factors; denominators $\neq 0$. For $l \neq j$ the factor $k = l$ vanishes; for $l = j$ all factors are 1.
2. Interpolation. For every $l = 0, \ldots, n$:

$$P_n(x_l) = \sum_{j=0}^{n} y_j \, \ell_j(x_l) = y_l.$$

3. Uniqueness. If $Q \in \prod_n$ also interpolates, then $D := P_n - Q \in \prod_n$ is 0 at all $n+1$ nodes. A nonzero polynomial of degree $\leqslant n$ has at most n zeros, so $D \equiv 0$.
4. V *nonsingular.* $Vc = 0$ means: coefficients c interpolate the data $y = 0$. So does 0; by 3., $c = 0$. ■

Example: Three Points, Two Ways

Example (*Data* $(0,1)$, $(1,3)$, $(2,2)$; $n=2$*). *Lagrange.*

$$\ell_0 = \tfrac{(x-1)(x-2)}{2}, \quad \ell_1 = \tfrac{x(x-2)}{-1}, \quad \ell_2 = \tfrac{x(x-1)}{2}, \qquad P_2 = 1\cdot\ell_0 + 3\,\ell_1 + 2\,\ell_2.$$

Expanding: $P_2(x) = 1 + \frac{7}{2}x - \frac{3}{2}x^2$.
Vandermonde.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \Rightarrow\ c_0 = 1,\ c_1 + c_2 = 2,\ 2c_1 + 4c_2 = 1 \ \Rightarrow\ c = \left(1, \tfrac{7}{2}, -\tfrac{3}{2}\right)^T.$$

Same polynomial, as the theorem says. (Check: $P_2(1) = 1 + 3.5 - 1.5 = 3$, $P_2(2) = 1 + 7 - 6 = 2$.)

Remark (*Which to use*). Lagrange needs no linear solve: the "system" in this basis is the identity, since $\ell_j(x_l) = \delta_{jl}$.

Vandermonde Systems in Practice

Remark (*Conditioning, computed in double precision*). Equally spaced nodes, $\kappa_2(V)$:

n	5	10	15	20
nodes in $[-1,1]$	$6.4 \cdot 10^1$	$1.4 \cdot 10^4$	$3.3 \cdot 10^6$	$8.3 \cdot 10^8$
nodes in $[0,1]$	$4.9 \cdot 10^3$	$1.2 \cdot 10^8$	$3.1 \cdot 10^{12}$	$8.9 \cdot 10^{16}$

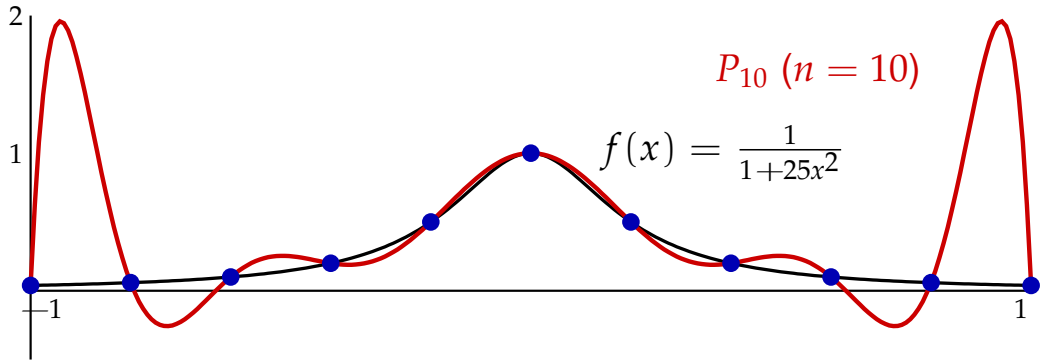
$\kappa_2(V)$ grows exponentially in n. By Lecture 15, the computed coefficients c can lose about $\log_{10} \kappa(V)$ digits: on $[0,1]$ with $n = 20$, possibly all 16.

Remark (*Practical rules*).

- MATLAB `polyfit(x,y,n)` solves this system; for large n it warns that the polynomial is badly conditioned.
- Few points (say $n \leqslant 5$): any form is fine.
- Many points: do not use one high-degree polynomial on equally spaced nodes (next slide). Use splines (Lectures 17–18) or special nodes (Lecture 18).

Higher Degree Is Not Always Better

Example (*Runge's example, equally spaced nodes on* $[-1,1]$*).*



Interpolate $f(x) = 1/(1+25x^2)$ at $x_j = -1 + 2j/n$. Maximum error on $[-1,1]$ (computed):

n	5	10	20
$\max \lvert f - P_n \rvert$	0.43	1.92	59.8

Remark (*Preview*). The error grows with n, although f is infinitely differentiable. Why, and how special nodes fix it: the error formula in Lecture 18.

From One Polynomial to Splines

Remark (*Two problems with one global polynomial*).

- **Oscillation:** high degree on equally spaced nodes can oscillate wildly near the ends (Runge).
- **Global coupling:** changing one y_j changes P_n on all of $[x_0, x_n]$ (it adds $\delta y_j \, \ell_j$).

Remedy: low, fixed degree on each subinterval, glued together smoothly.

Definition (*Spline*). Let $a := x_0 < x_1 < \cdots < x_n =: b$, $n \geqslant 1$ (the *knots*), and $k \geqslant 2$. A function $S : [a, b] \to \mathbb{R}$ is a *spline of order* k (degree $k-1$) if

(a) $S \in C^{k-2}[a, b]$ ($S, S', \ldots, S^{(k-2)}$ exist and are continuous on $[a, b]$; for $k = 2$: S continuous), and

(b) on each $[x_{j-1}, x_j]$, $j = 1, \ldots, n$, S coincides with a polynomial in $\prod_{k-1}$.

Remark. $k = 2$: piecewise linear ("connect the dots"). $k = 4$: cubic spline, the workhorse (MATLAB `spline`).

Looking Ahead

- **Next lecture:** Splines